ECE 4180 Final Project Proposal – One Proposal per Team

Team member 1: ___Zachary Seletsky___________________ Section: ______B______

Team member 2: ____ Section: ____________

Please answer the following questions about your final project:

What microcontroller and peripherals are you going to be using? Explain what each peripheral is going to be doing. If you need to order a part, please include it here along with a link of what needs to be purchased.

I plan on creating a toy RC car and RC controller using the mbed nxp1768 and an Arduino Due, which is Cortex M3 based. I'll be using the Due because I happen to have one and it uses an M3 chip. I will not need any parts ordered.

In Scope of project:

The RC car will use the Arduino Due, two standard motors and a motor driver (TB6612FNG) for the powertrain, a servo (HS-422) for steering, an ultrasonic sensor (HC-SR04) for collision avoidance, and a Bluetooth module (HC01) for communication with the RC controller.

The RC controller will use the mbed nxp 1768, a rotary pulse generator for steering input, a button for throttle, a button for braking, a switch for changing gears (forward-reverse), a capacitive keyboard (MPR121) for unlocking the controller upon powering it on, a uLCD screen (uLCD-144G2for conveying information such as battery status, Bluetooth status, for a phone-style unlock screen, and a Bluetooth module (HC01) for communication with the RC car.

Stretch Scope of project (aka will do if time permits):

The RC car may have a speaker (tbd) with an audio amp for playing sounds transmitted from the controller and an IMU (ICM-20948) for tracking inertial data such as acceleration and incline.

The RC controller may have a micropohone (SPW2430) for live transmitting sound from the controller to the RC car's speaker and the uLCD screen may also support a visualization of the IMU data (possibly like g-force sensors on track focused cars, example below).

(pictured: g-force sensor on a toyota 86)

What is the purpose of your final project? Does it solve a specific problem? Make a user's day more convenient? Or is it a game for people to play?

It's a game for the user's entertainment.

Please provide any similar embedded systems/solutions that exist and how yours is different or improves upon it.

There are lots of RC car models, but most don't have a collision avoidance system, a speaker/mic combo, a dedicated braking input and forward/reverse selector (most have a single input for braking + reverse) or any kind of screen on the controller. Some examples from amazon linked below.

https://a.co/d/7O6spre

https://a.co/d/4P7RzwZ